



UNIT IV

Role of Science & Technology in Disaster Management



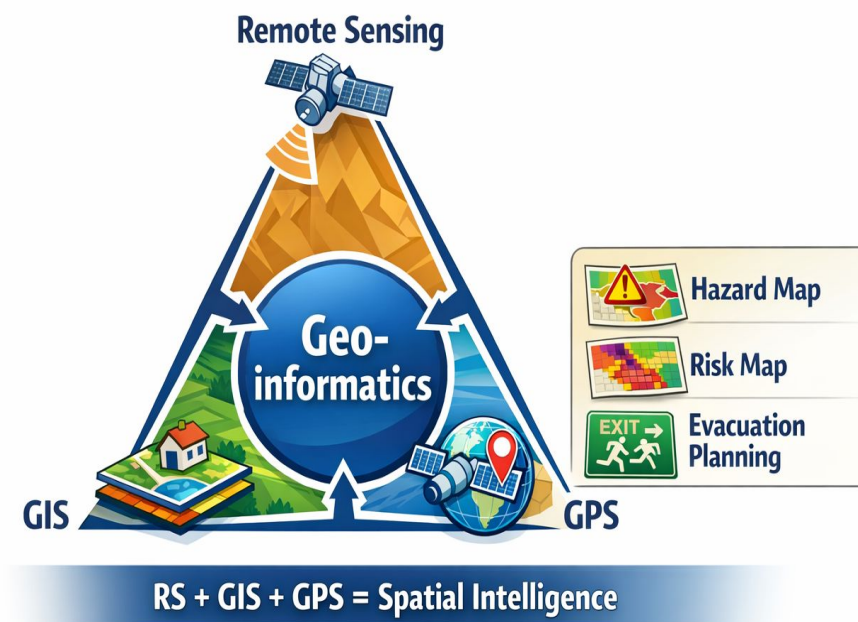
- Science and Technology (S&T) play a critical role in **all phases of disaster management** — prevention, preparedness, response, and recovery.
- Modern tools such as **Remote Sensing, GIS, GPS, Early Warning Systems, and Structural Engineering** enable evidence-based decision-making.
- S&T reduces disaster impact by improving **risk identification, early warning, planning, and resilient construction**.
- Integration of technology ensures **faster response, better coordination, and long-term risk reduction**



Geo-informatics — Concept and Importance

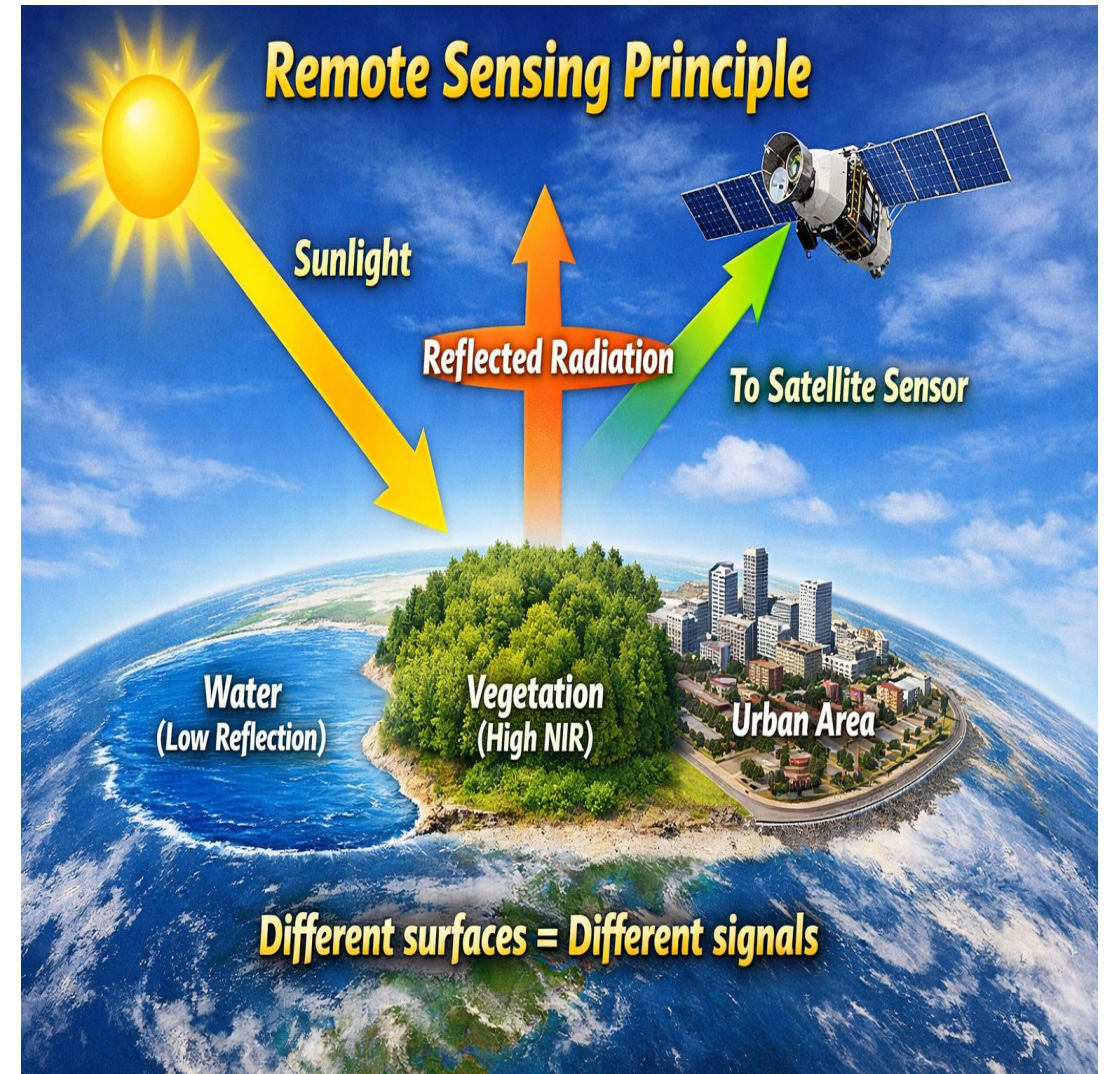


- Geo-informatics is the integration of **Remote Sensing (RS)**, **Geographic Information System (GIS)**, and **GPS**.
- It provides **location-based intelligence** essential for disaster management.
- Disasters are spatial events; geo-informatics helps identify **where, how, and whom they affect**.
- It enables **accurate mapping, monitoring, and decision-making**



Remote Sensing — Concept

- Remote sensing is the science of **collecting information about Earth without direct contact**.
- It uses satellites and sensors to detect **reflected or emitted electromagnetic radiation**.
- Different surfaces have unique **spectral signatures**.
- It provides **real-time and large-scale data** for disaster management.

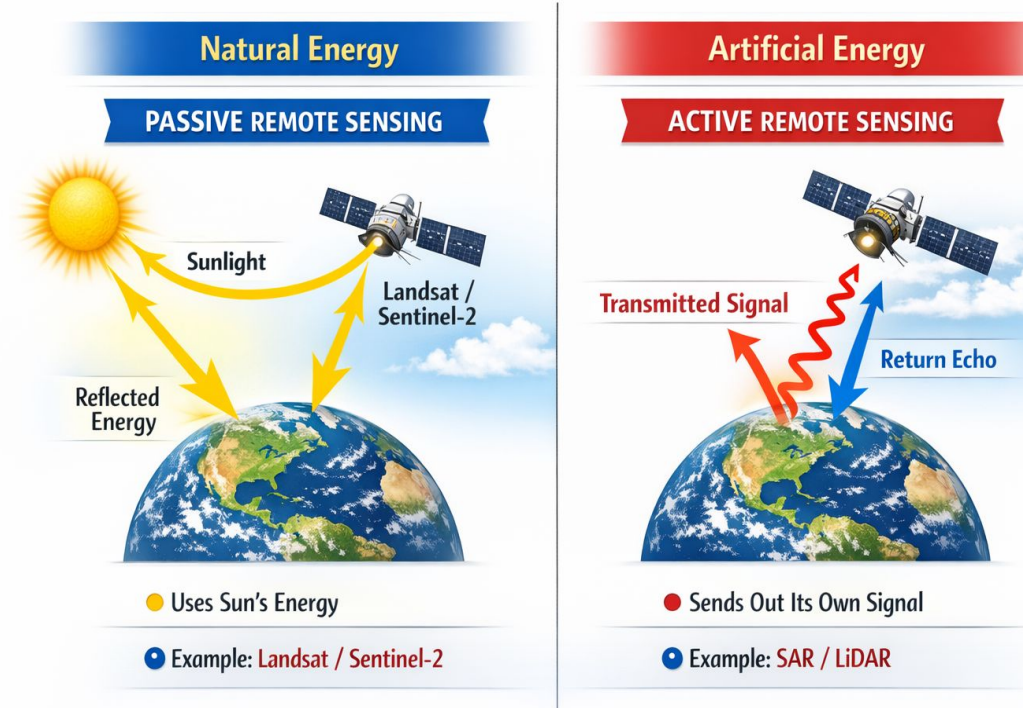


Types of Remote Sensing



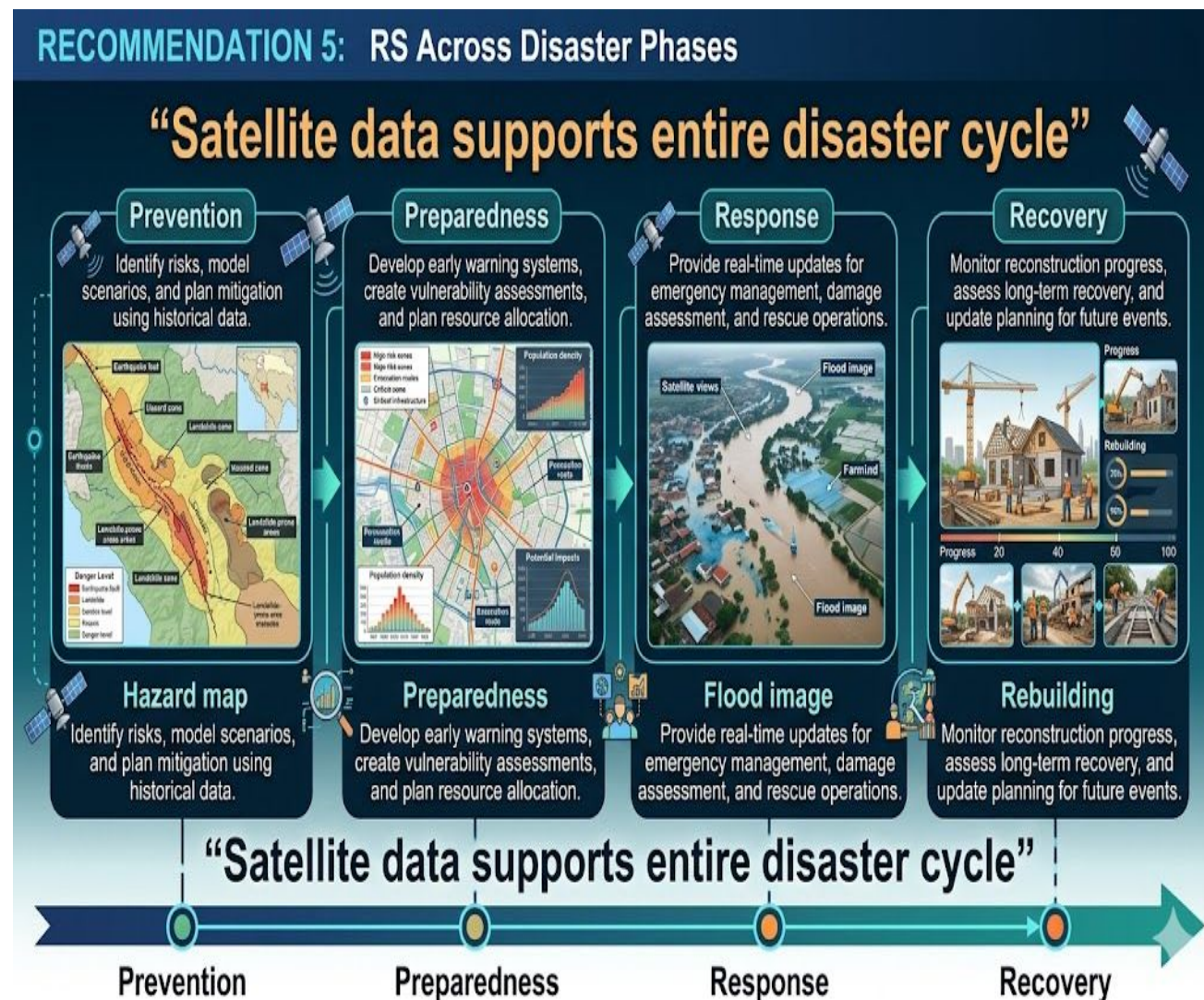
- Remote sensing is classified into **Passive and Active systems**.
- Passive sensors detect **natural radiation (sunlight/thermal)**.
- Active sensors emit their own signal and measure the **returned signal**.
- SAR and LiDAR are important active systems used in disasters

Passive vs Active Remote Sensing



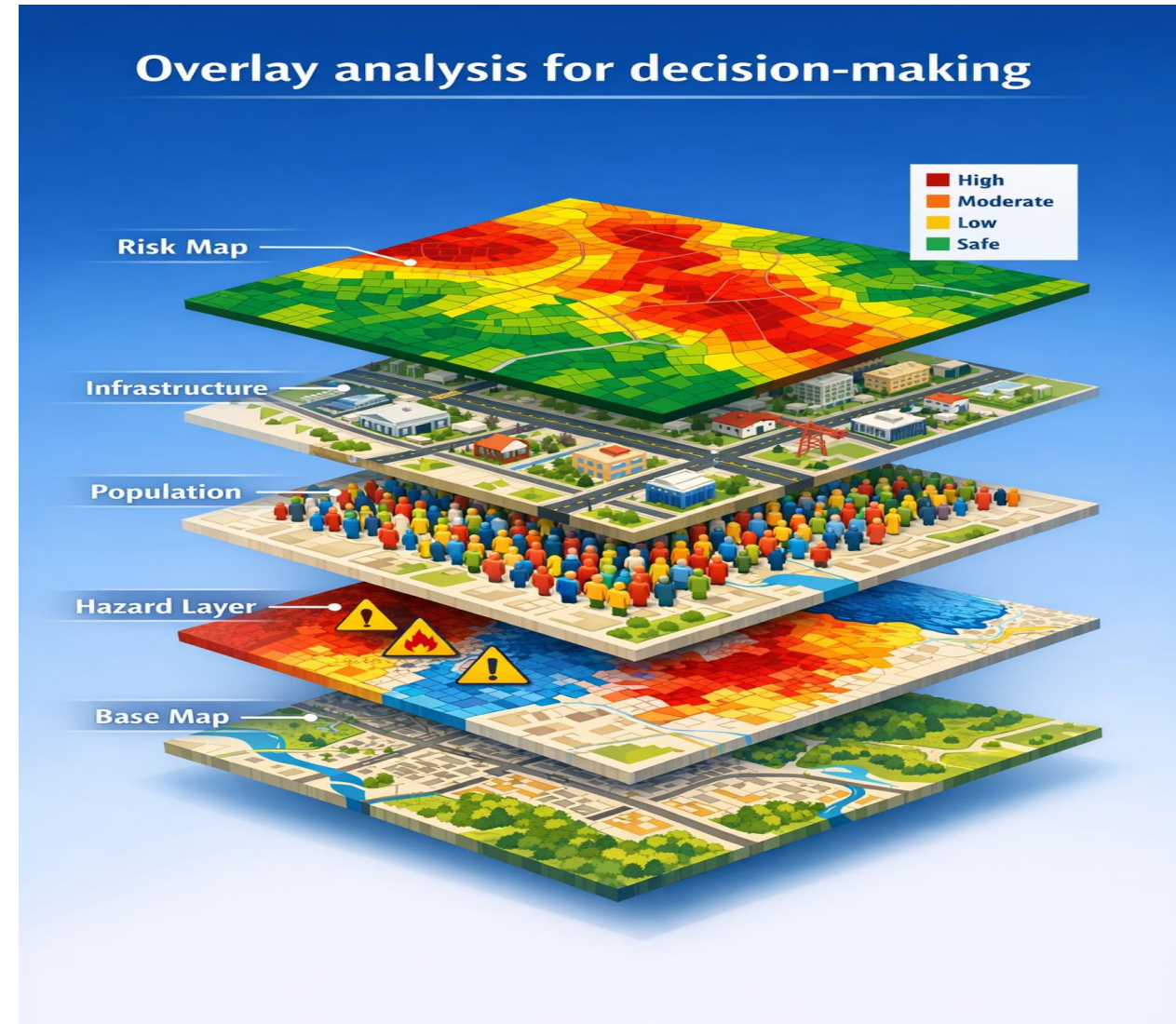
Applications of Remote Sensing in Disaster Management

- Remote sensing is used in all disaster phases.
- Pre-disaster: Hazard mapping and vulnerability assessment.
- During disaster: Real-time damage and flood mapping.
- Post-disaster: Recovery monitoring and loss estimation.
- It provides **rapid and large-scale assessment**.



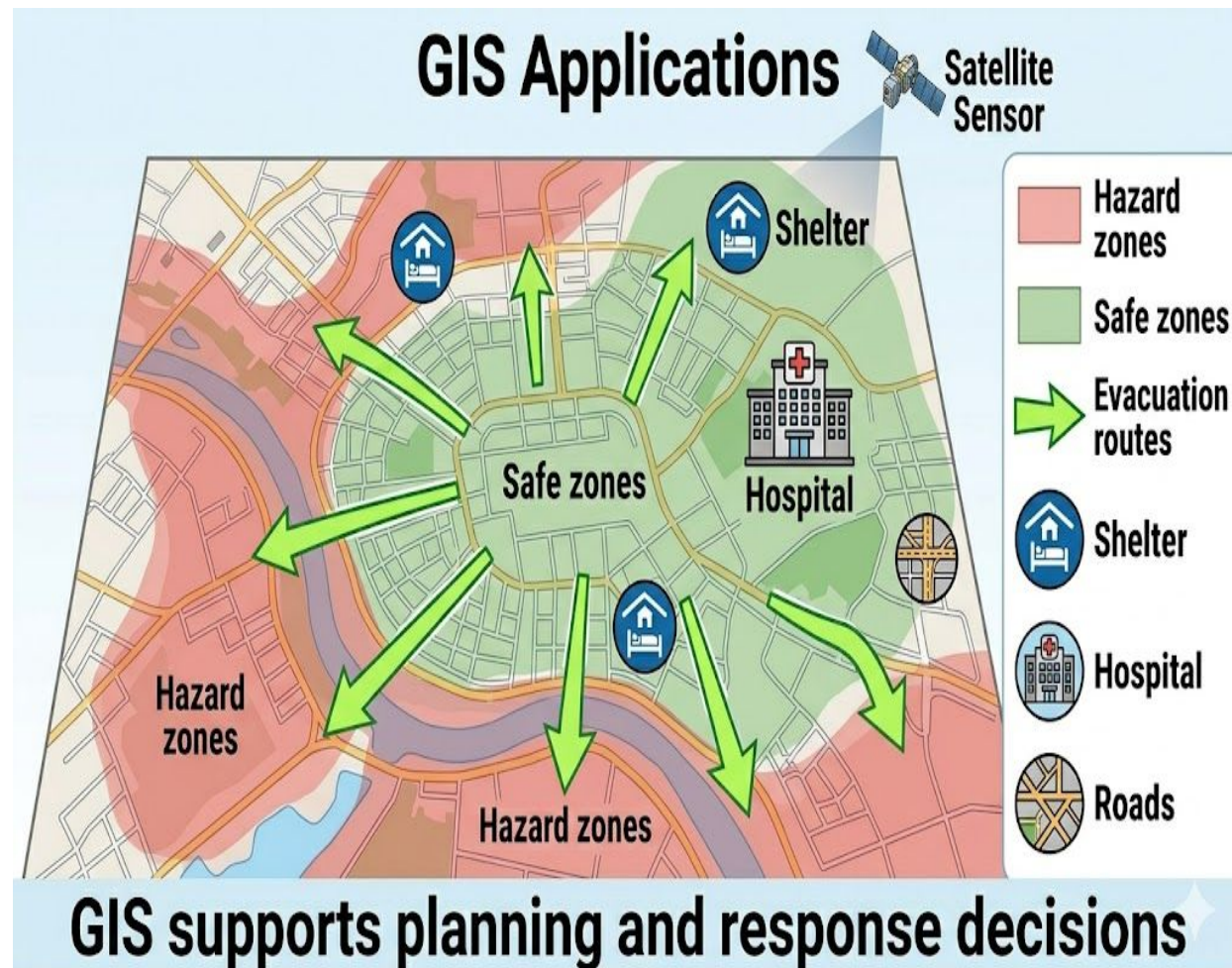
Geographic Information System (GIS)

GIS is a system for **capturing, storing, analysing, and displaying geographic data**. It integrates spatial data with **attribute data (population, infrastructure)**. It allows **multi-layer analysis and decision-making**. GIS supports evacuation planning, risk mapping, and damage assessment.



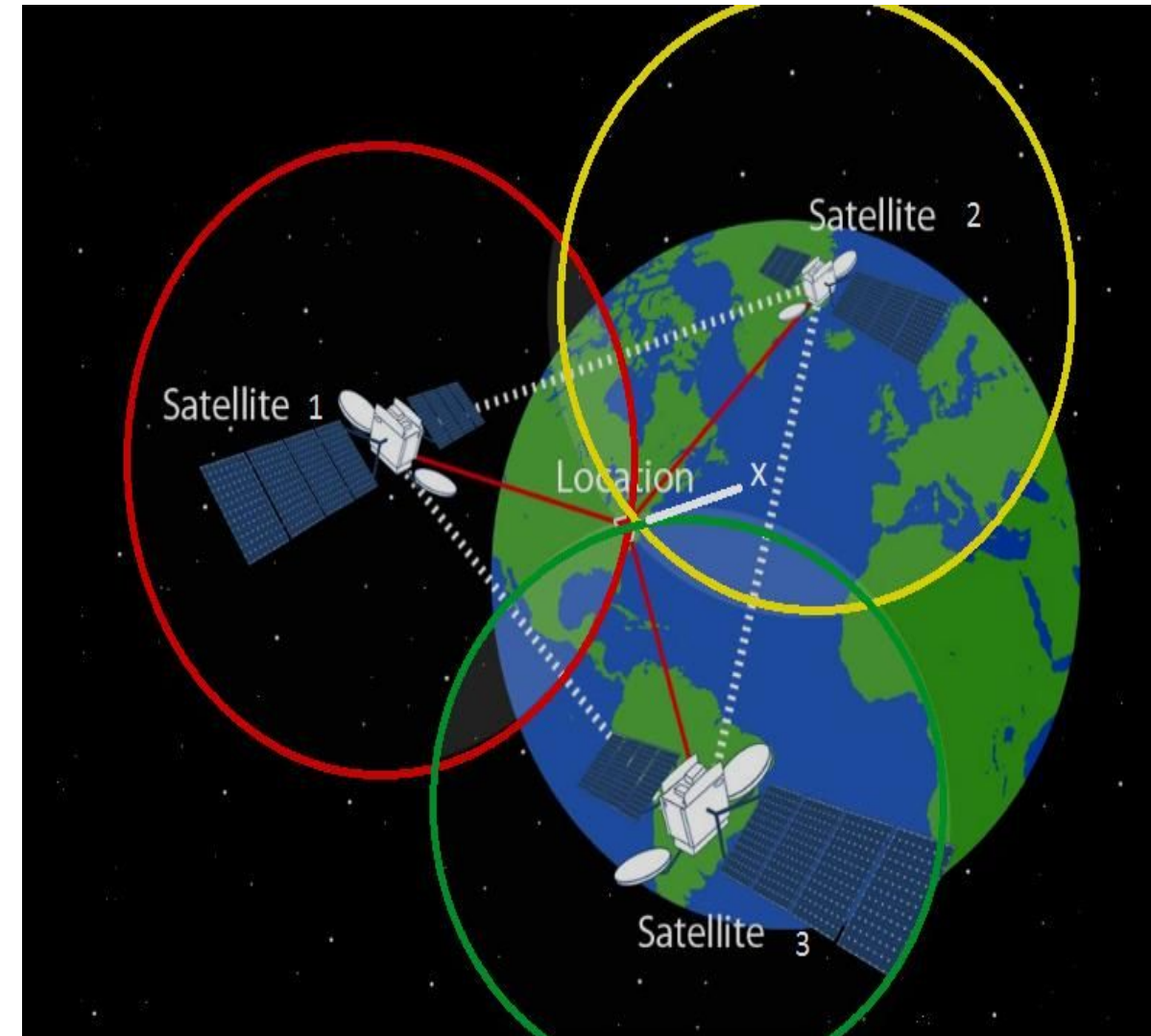
GIS Applications in Disaster Management

- GIS supports hazard mapping, evacuation planning, and response coordination.
- It helps identify **high-risk zones and vulnerable populations**.
- It enables **route optimization and resource allocation**.
- Used widely by **NDMA, NRSC, and state agencies**.



Global Positioning System (GPS)

- GPS provides **accurate location, velocity, and time information.**
- It works using **satellites and receivers.**
- It supports field operations such as **rescue, tracking, and mapping.**
- India's NavIC system provides **regional positioning accuracy.**

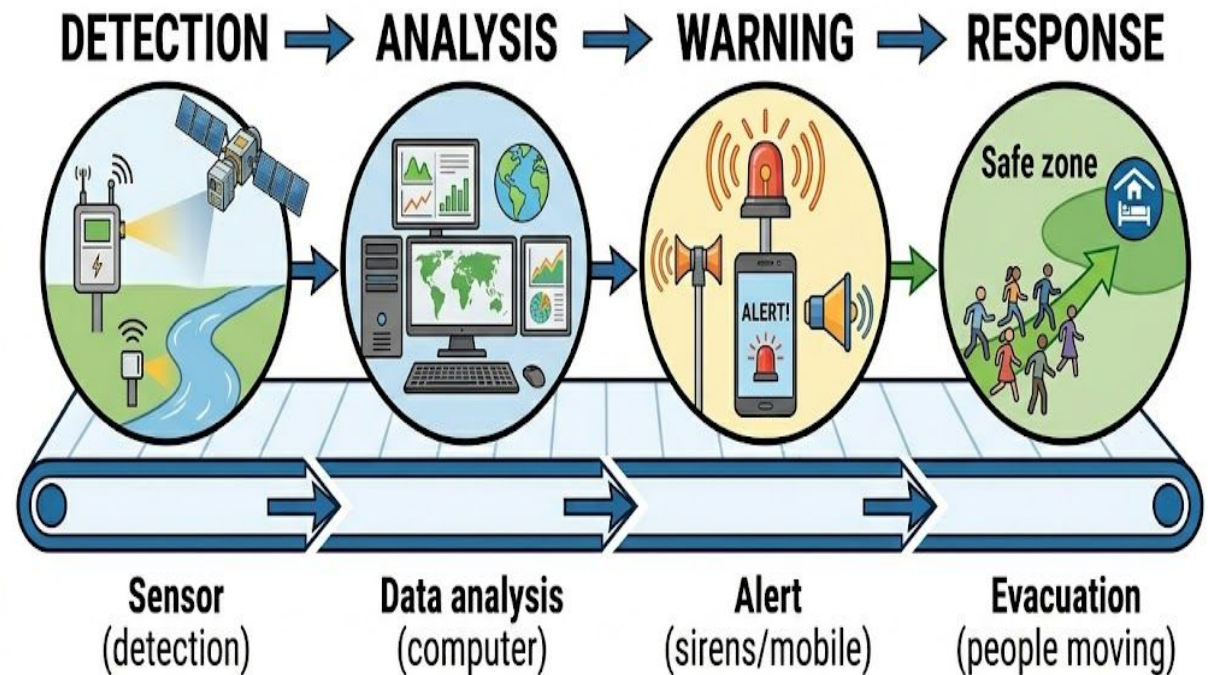


Early Warning System (EWS) – Concept

An Early Warning System (EWS) is an integrated system that **detects hazards, analyses risk, and provides timely warning before impact.**

- It helps communities take **preventive and preparedness actions.**
- EWS reduces **loss of life and property** significantly.
- It is effective only when warnings are **timely, accurate, and actionable.**

EARLY WARNING SYSTEM FLOW



EARLY WARNING SAVES LIVES BEFORE DISASTER STRIKES

Components of Early Warning System

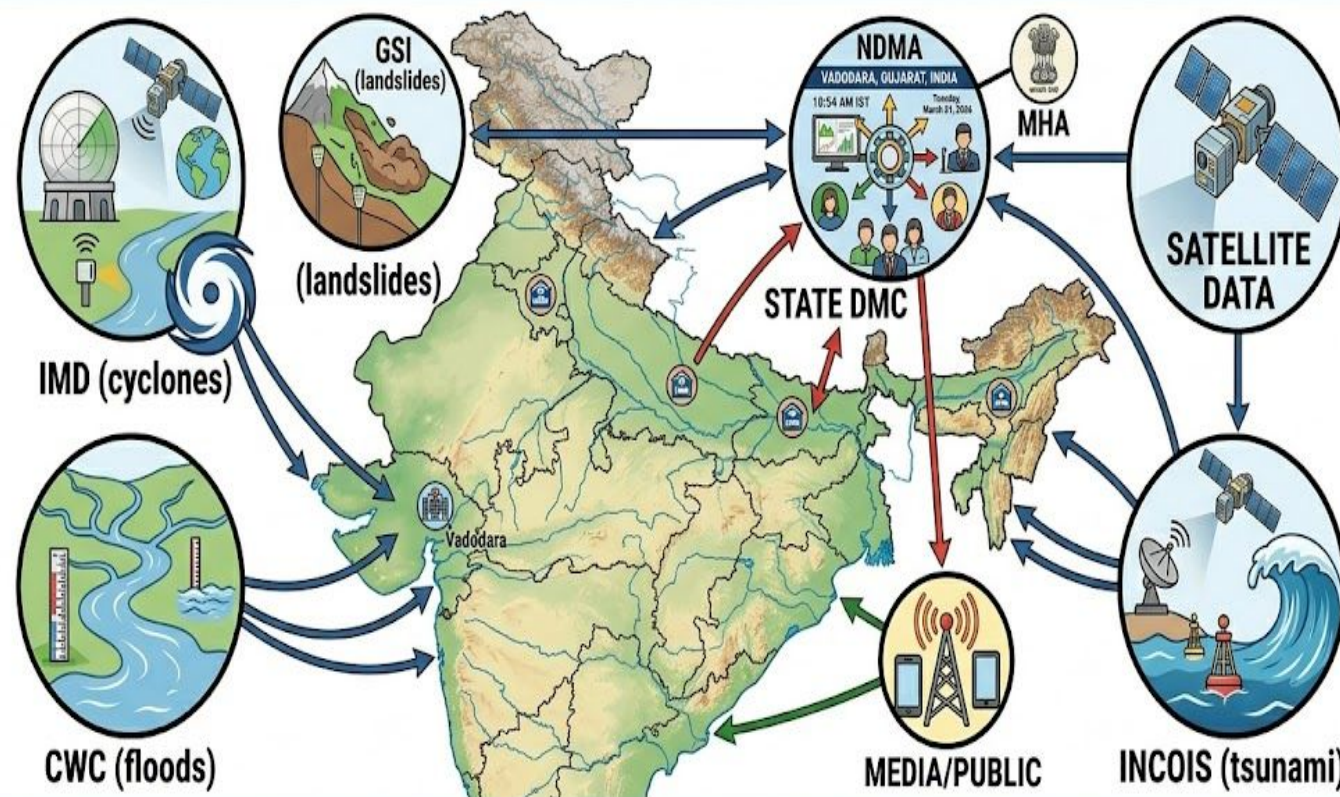
- EWS consists of four key components.
- Risk knowledge includes understanding hazards and vulnerable areas.
- Monitoring systems continuously track hazard indicators.
- Communication ensures warnings reach people effectively.
- Response capability ensures people take appropriate action.



Hazard-Specific Early Warning Systems in India

- India has developed hazard-specific warning systems.
- IMD provides cyclone and weather warnings.
- CWC provides flood forecasting.
- INCOIS provides tsunami warnings.
- IMD and GSI support landslide and earthquake monitoring.

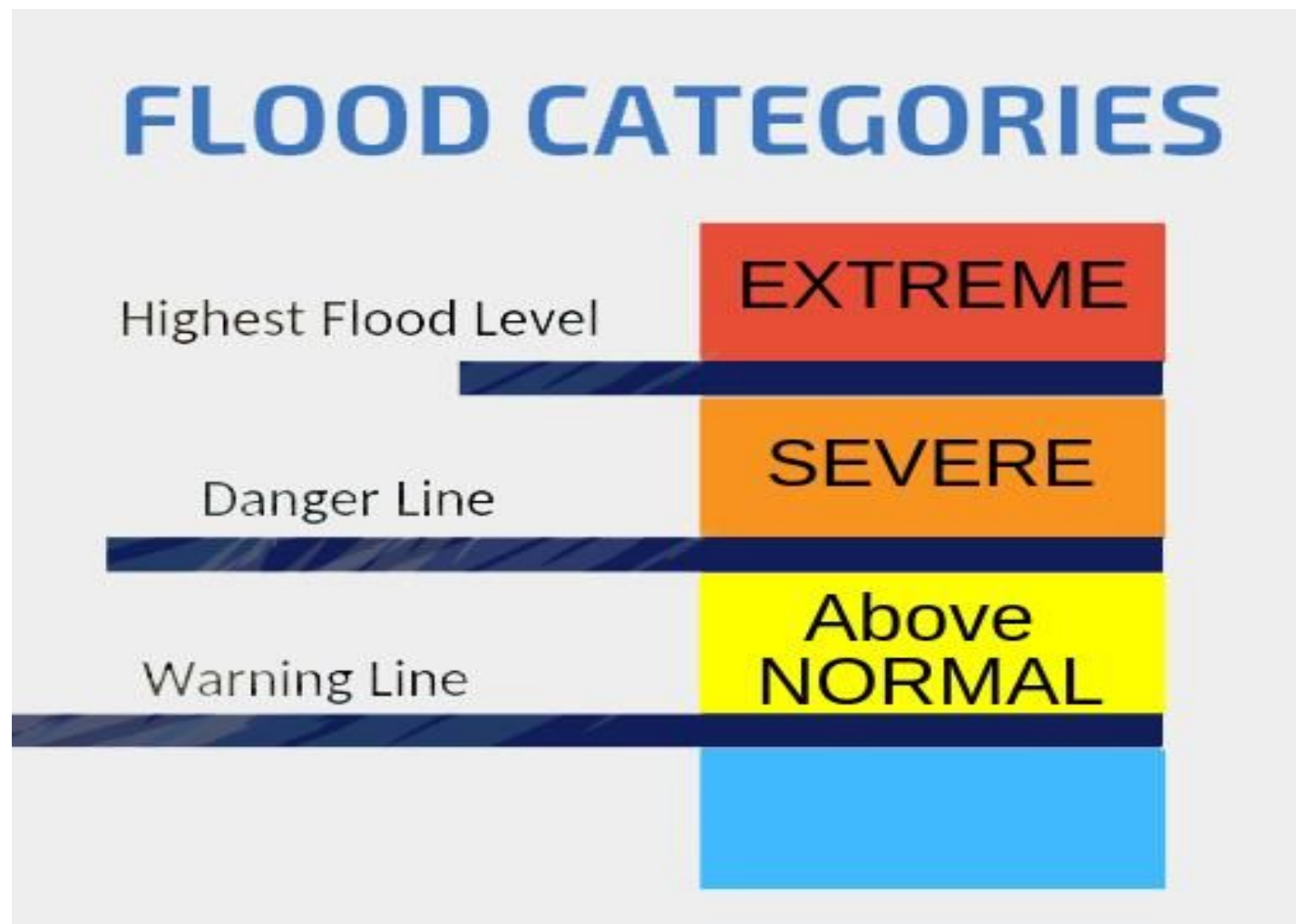
INDIA EARLY WARNING SYSTEM (EWS) MAP



INTEGRATED SYSTEMS, PRECISE WARNINGS, SAVING LIVES

Cyclone and Flood Warning Systems

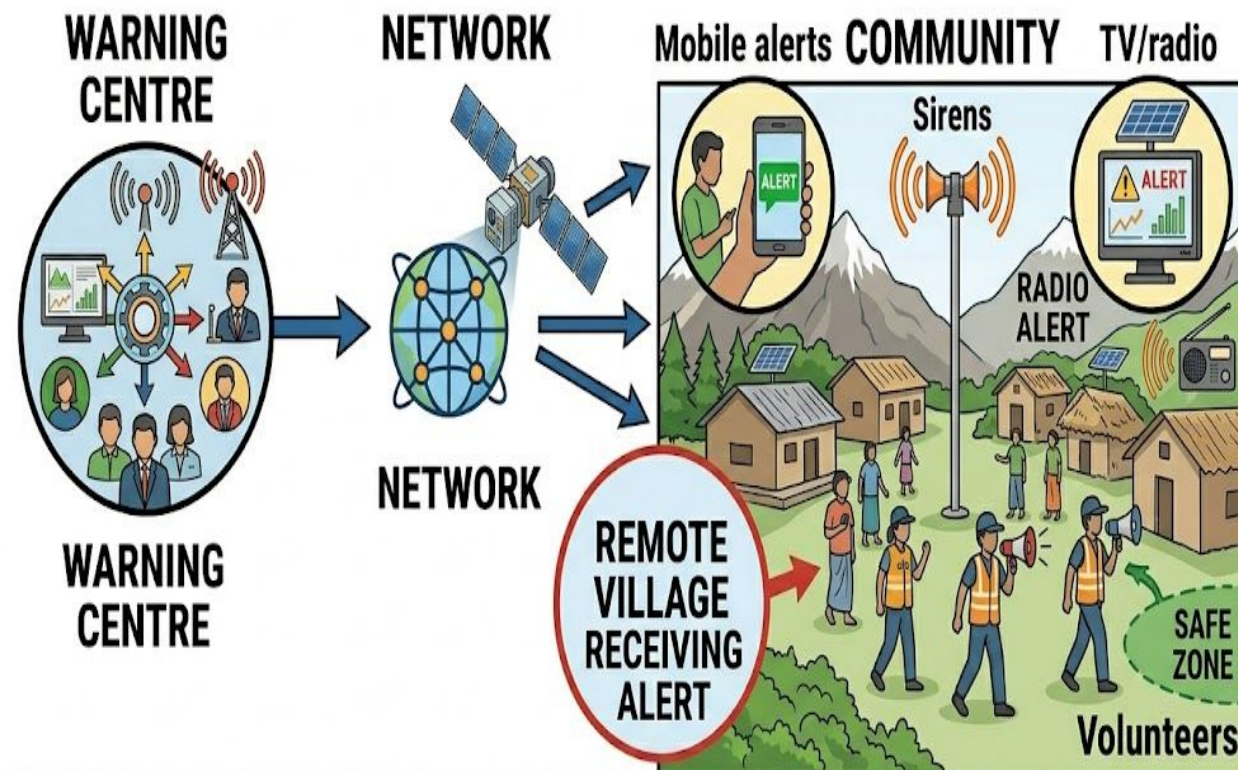
- IMD provides cyclone warnings with **5-day forecasts and colour-coded alerts.**
- CWC provides flood forecasts with **24–72 hour lead time.**
- Warnings include **intensity, location, and expected impact.**
- These systems enable **timely evacuation and preparedness**



Last-Mile Warning Dissemination

- The biggest challenge is ensuring warnings reach **every individual at risk**.
- Multiple channels are used: SMS, TV, radio, sirens, and social media.
- Cell Broadcast sends alerts to **all phones in a specific area**.
- Community volunteers ensure **last-mile communication**.

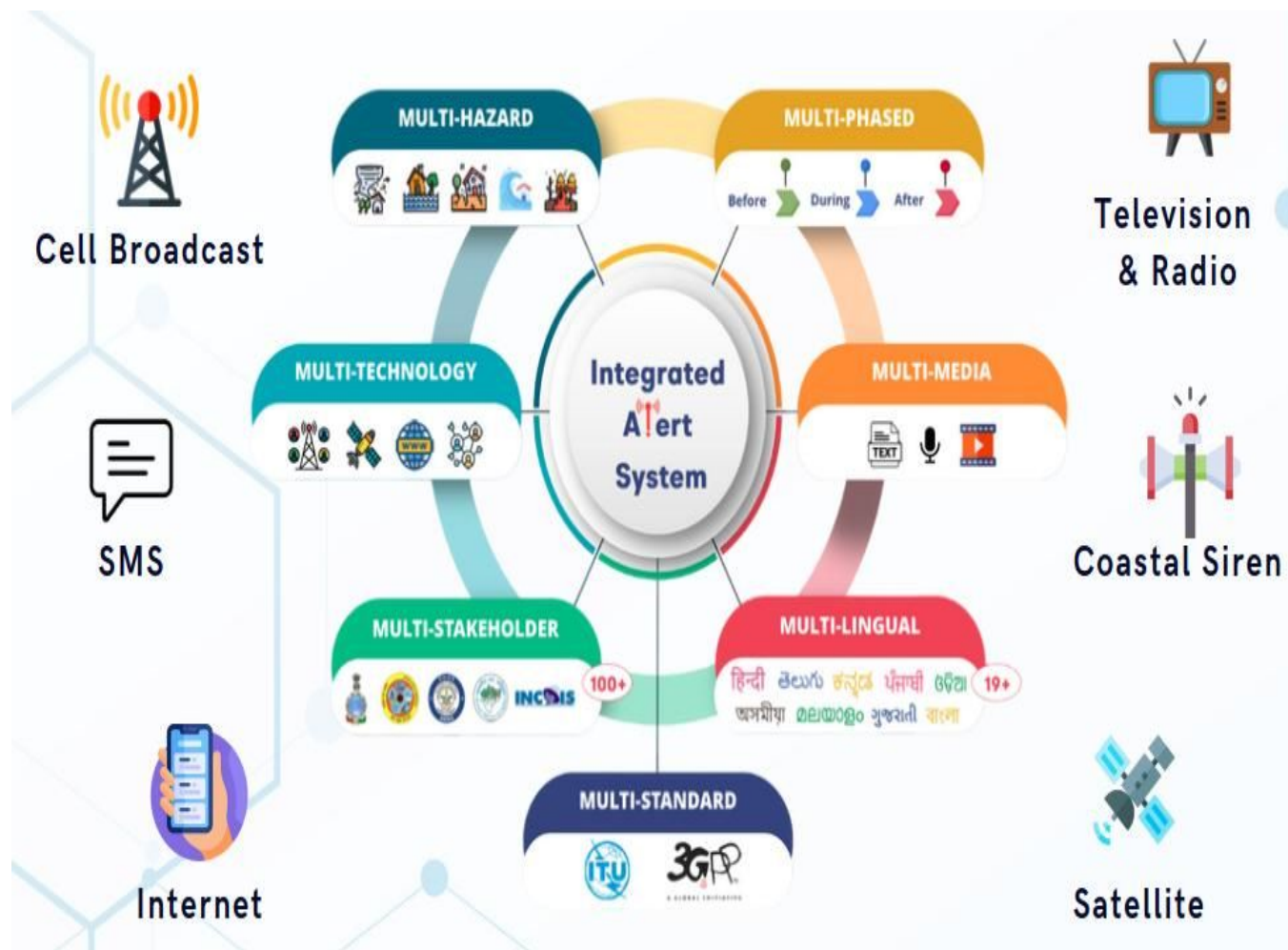
ALERT DISSEMINATION FLOW & CHANNELS



WARNING MUST REACH THE LAST PERSON

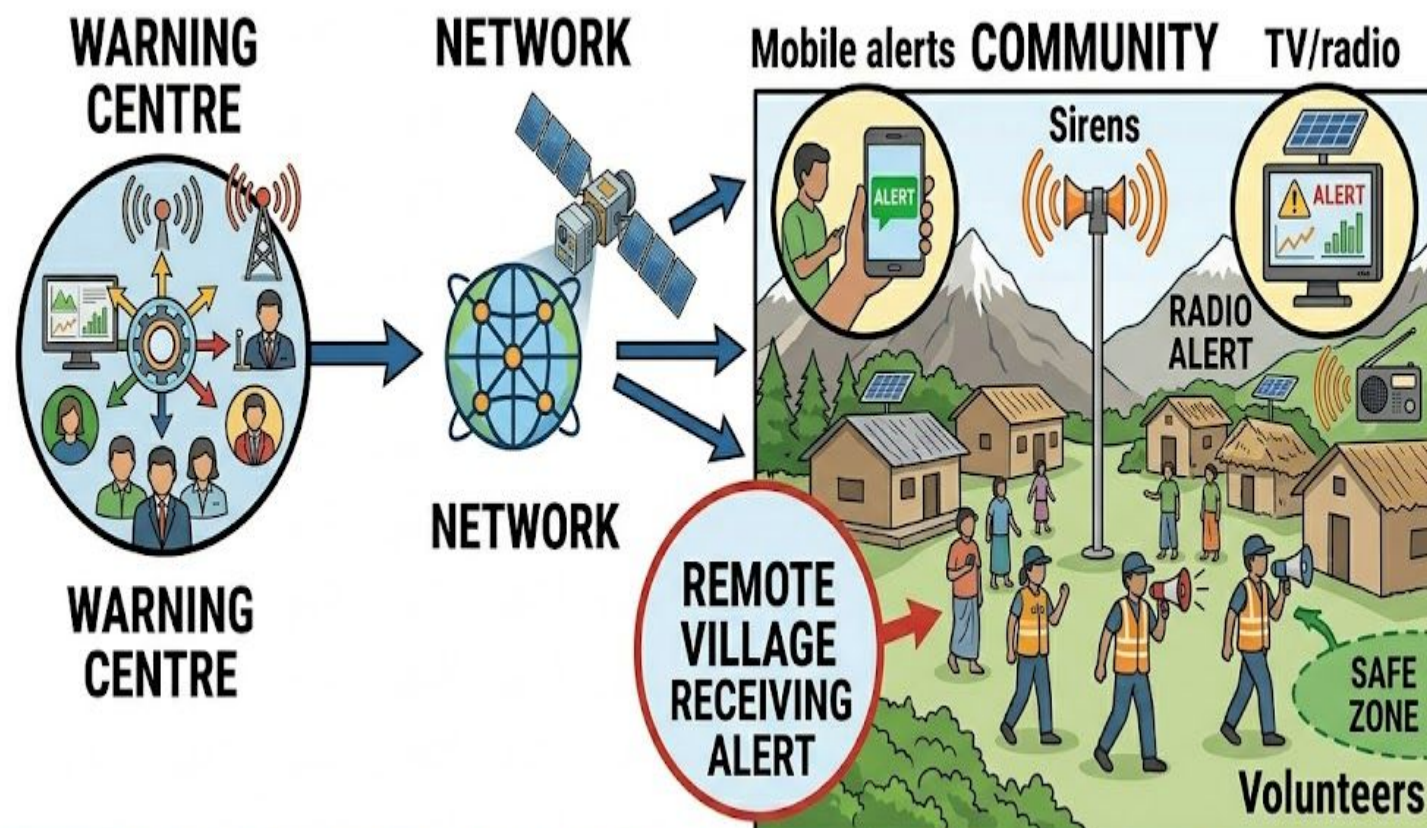
Common Alerting Protocol (CAP)

- CAP is a standard system for **sending warning messages across multiple platforms.**
- One message can be delivered through **SMS, TV, radio, and apps simultaneously.**
- It ensures **fast and uniform communication.**
- CAP improves **efficiency and coordination.**²²



- Reliable communication is critical during disasters.
- Systems include **NICNET, VSAT, satellite phones, and HAM radio.**
- These systems work even when **normal networks fail.**
- They ensure coordination between **response agencies.**

ALERT DISSEMINATION FLOW & CHANNELS



WARNING MUST REACH THE LAST PERSON

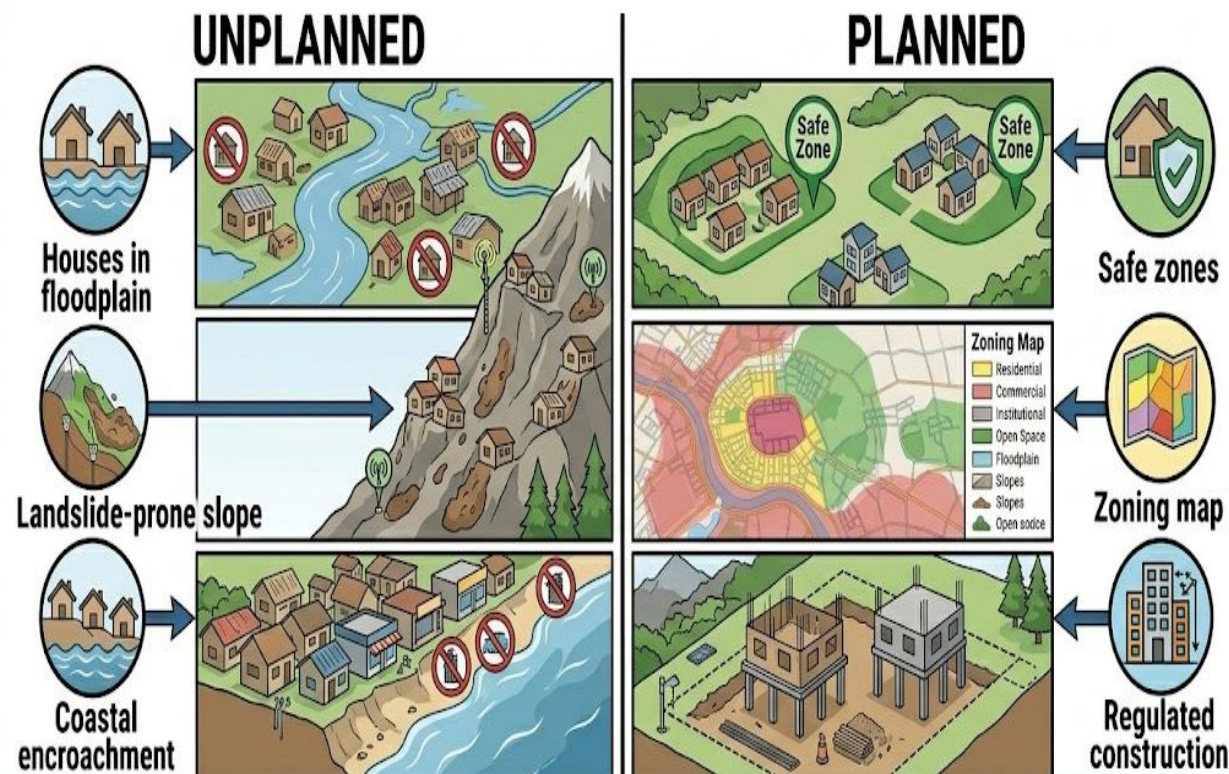
- Communication enables **coordination, resource allocation, and decision-making.**
- It connects **command centres, field teams, and affected communities.**
- Timely communication reduces **confusion and delays.**
- It is essential for **effective disaster response operations.**



Land Use Planning — Role in Disaster Management

- Land use planning controls **where and how development occurs**.
- Disaster risk increases when people settle in **hazard-prone areas** such as floodplains and coastal zones.
- Proper planning helps in **preventing risk rather than reacting to disasters**.
- It is the most **cost-effective disaster risk reduction strategy**

PLANNED VS UNPLANNED DEVELOPMENT



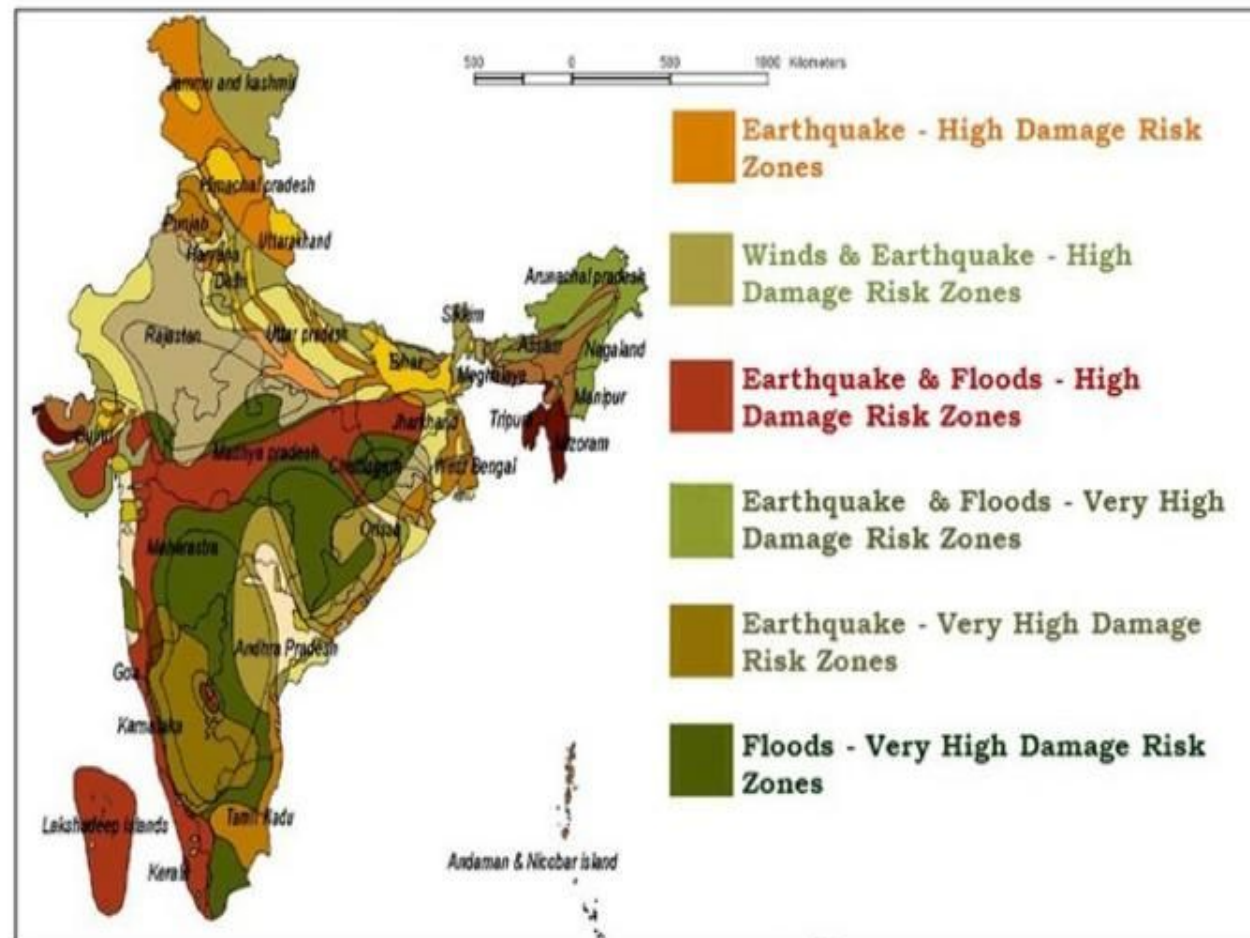
UNPLANNED DEVELOPMENT vs. UNPLANNED DEVELOPMENT

Hazard Zonation — Scientific Basis



- Hazard zonation divides areas based on **level of risk exposure**.
- It helps in planning safe construction and development.
- Examples include **seismic zones, flood zones, and coastal zones**.
- It provides a **scientific basis for land use planning**.

Showing Multi Hazard Prone Areas of India



Source: Vulnerability Atlas Building Material and Technology Promotion Council (BMTPC).

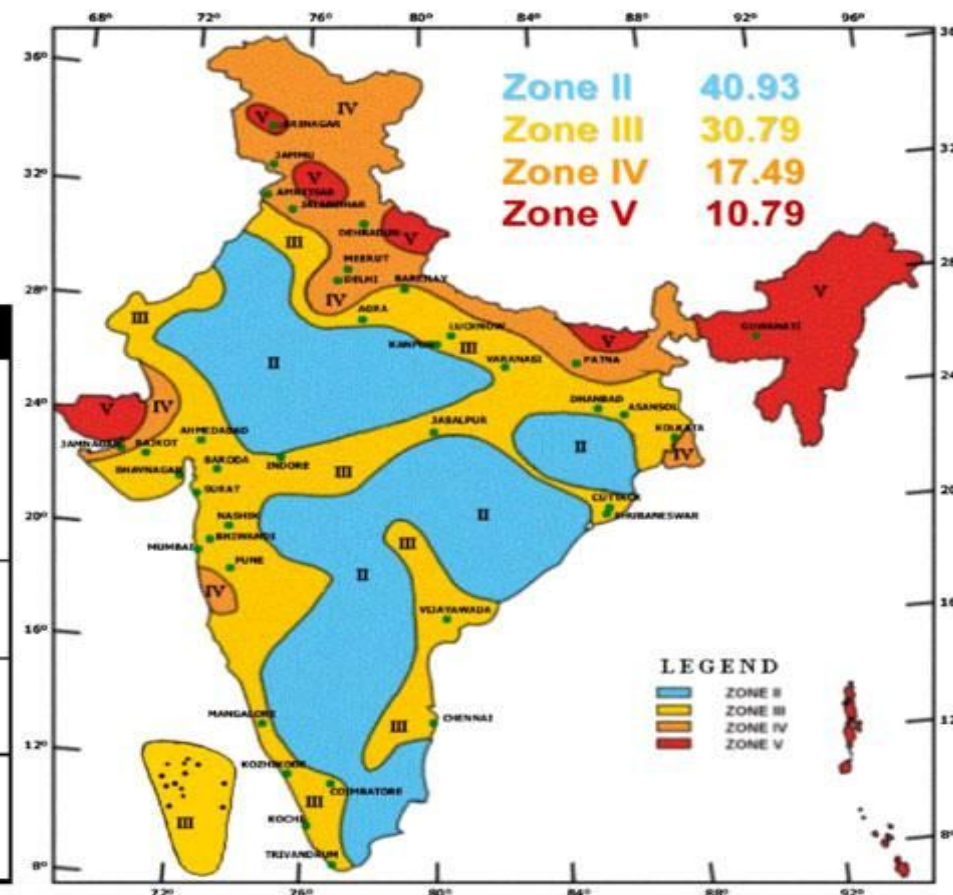
Seismic Zonation in India

- India is divided into **four seismic zones (II to V)** based on earthquake risk.
- Zone V represents the **highest risk areas**.
- Zone II represents the **lowest risk areas**.
- Building design standards vary according to zone.

Seismic Zone Map of India: -2002

About 59 percent of the land area of India is liable to seismic hazard damage

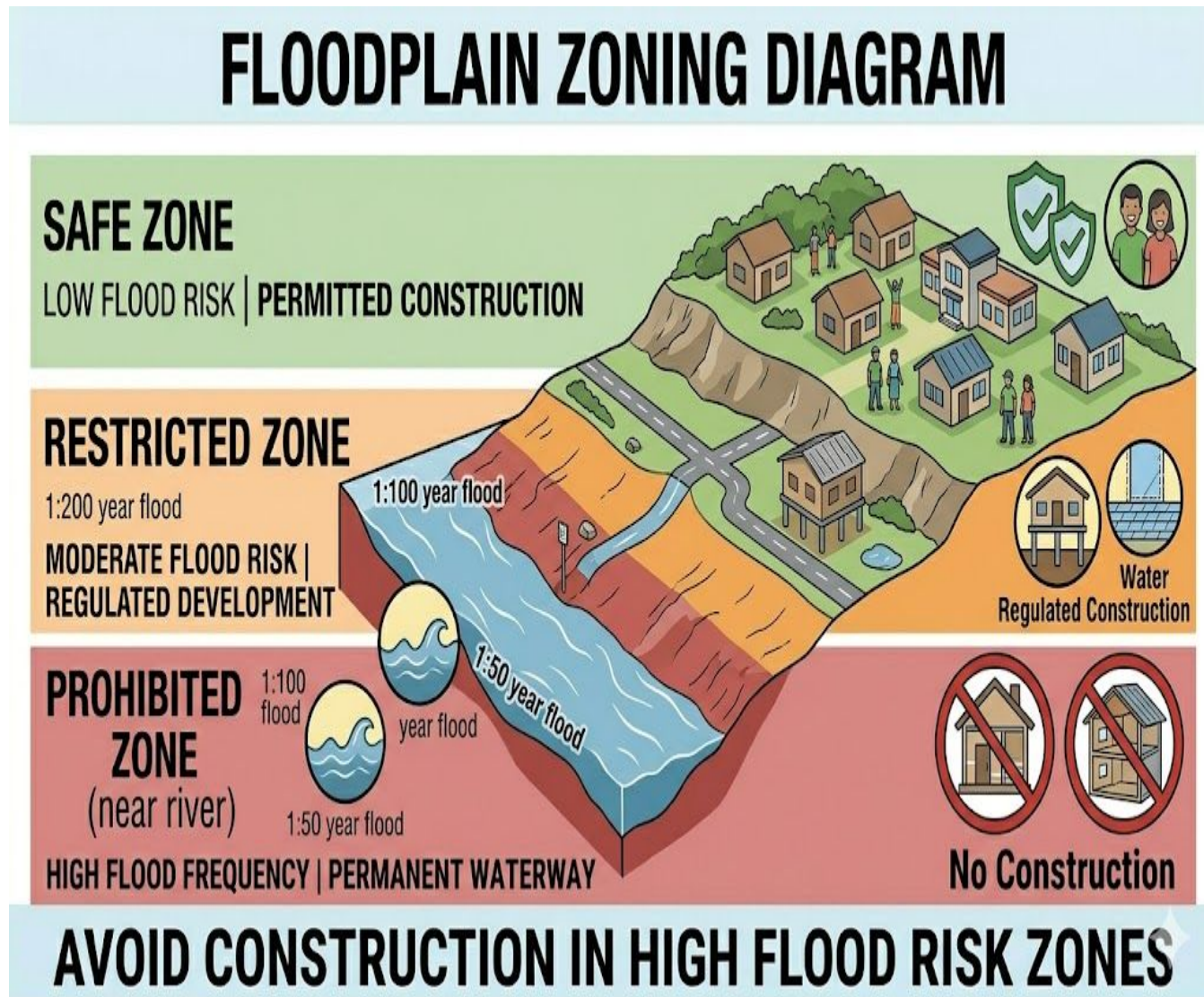
Zone	Intensity
Zone V	Very High Risk Zone Area liable to shaking Intensity IX (and above)
Zone IV	High Risk Zone Intensity VIII
Zone III	Moderate Risk Zone Intensity VII
Zone II	Low Risk Zone VI (and lower)



Seismic zonation and intensity map of India

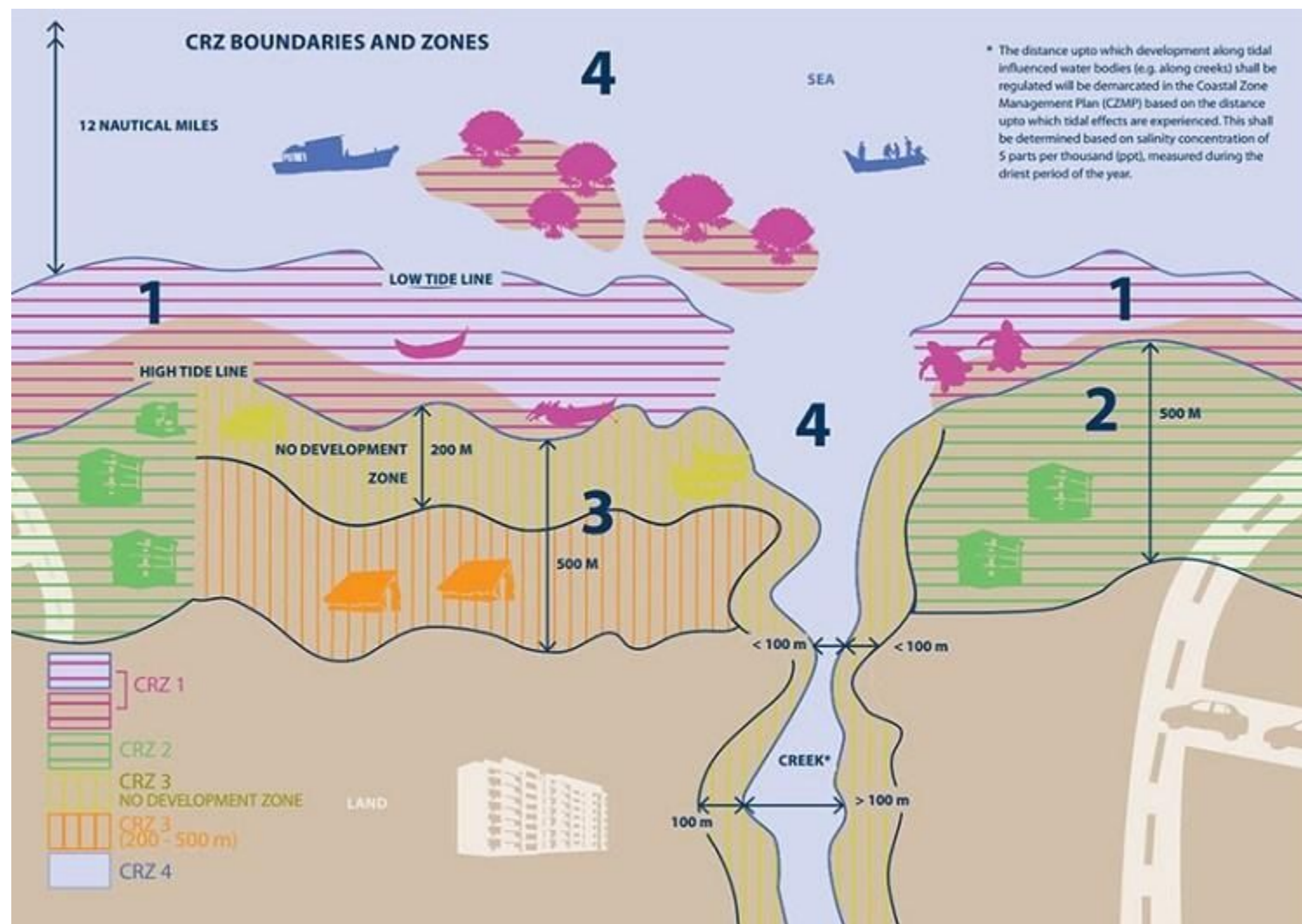
Flood Plain Zoning

- Floodplain zoning regulates construction in **areas prone to flooding**.
- It divides land into **prohibited, restricted, and safe zones**.
- Prohibited zones allow **no construction**.
- Restricted zones allow **controlled development**.
- It reduces flood damage and loss.



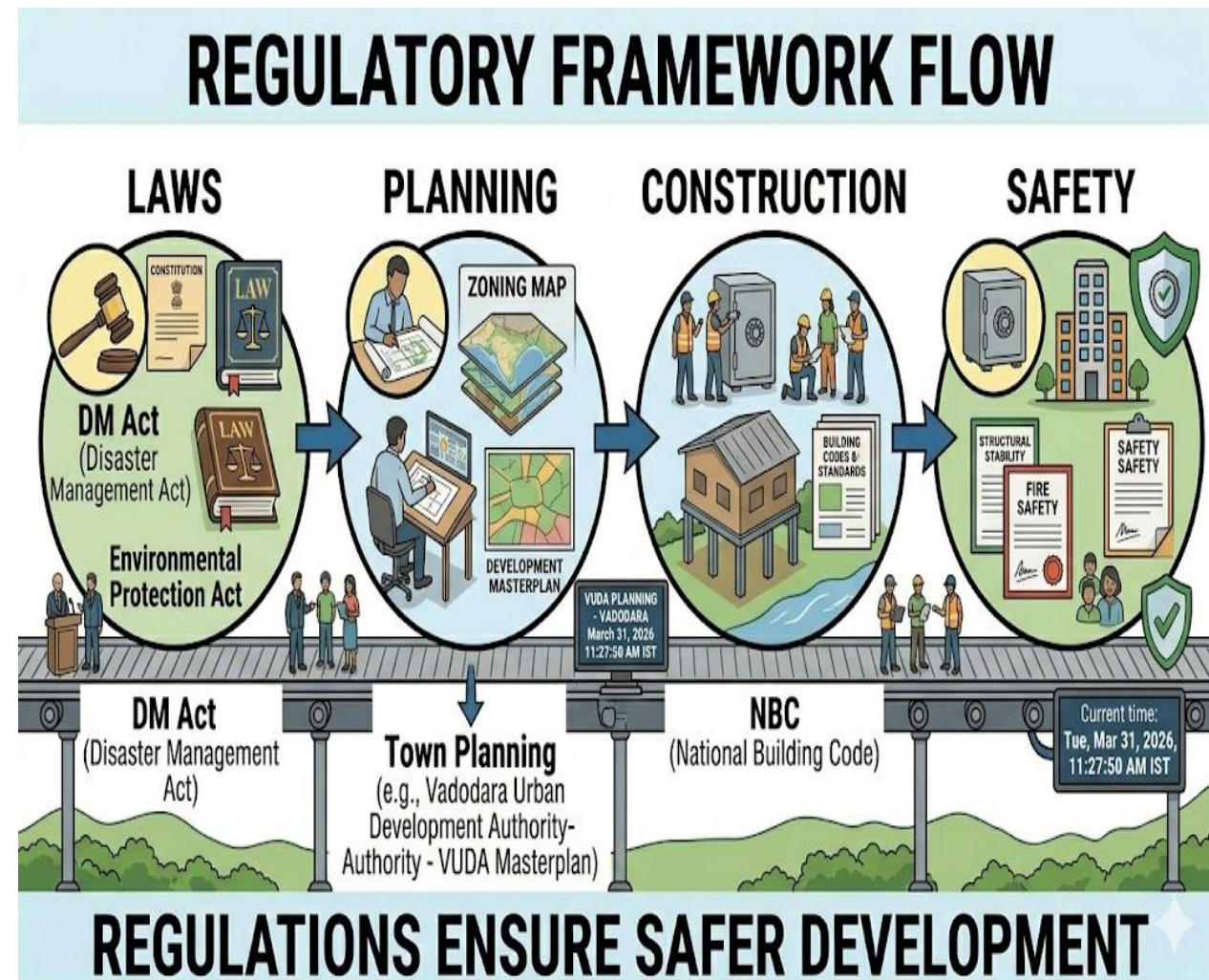
Coastal Regulation Zone (CRZ)

- CRZ regulates development in **coastal areas**.
- It classifies areas into **CRZ-I, II, III, and IV**.
- Sensitive areas have **strict restrictions on construction**.
- It helps in reducing **cyclone and tsunami impact**.



Regulatory Framework in India

- Disaster management regulations are supported by various laws.
- Key frameworks include **DM Act 2005, NBC 2016, and Building Bye-laws.**
- Town planning acts regulate **land use and development.**
- EIA ensures environmental and disaster risk considerations



Challenges in Land Use Planning



- Hazard maps are not always integrated into planning systems.
- Encroachment in hazard zones increases risk.
- Multiple authorities create coordination issues.
- Rapid urbanisation leads to unplanned development.



Importance of Risk-Sensitive Land Use Planning



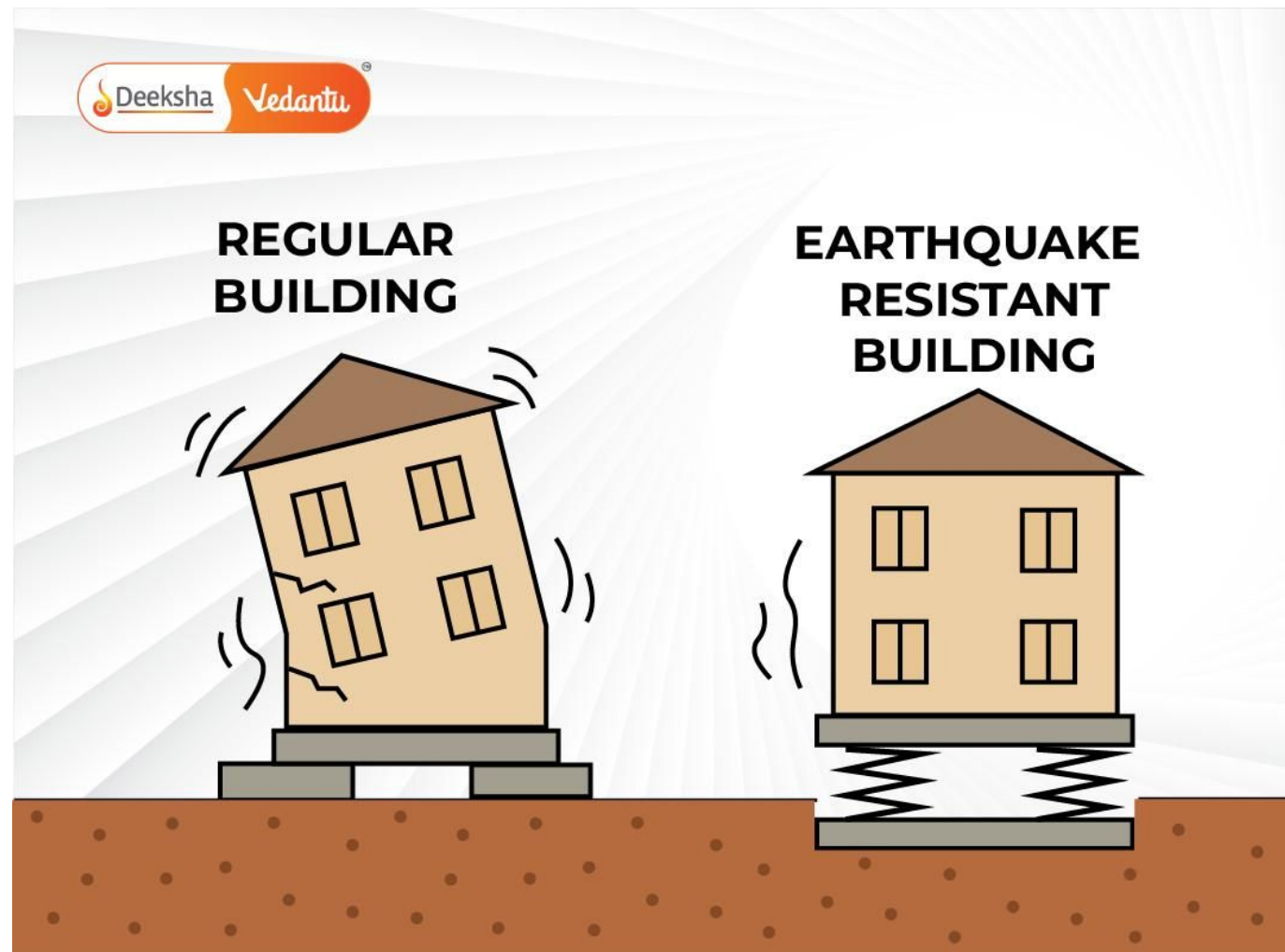
- Risk-sensitive planning reduces **future disaster losses**.
- It ensures safe location of **infrastructure and settlements**.
- It supports sustainable and resilient development.
- It is a key component of **disaster risk reduction strategies**.



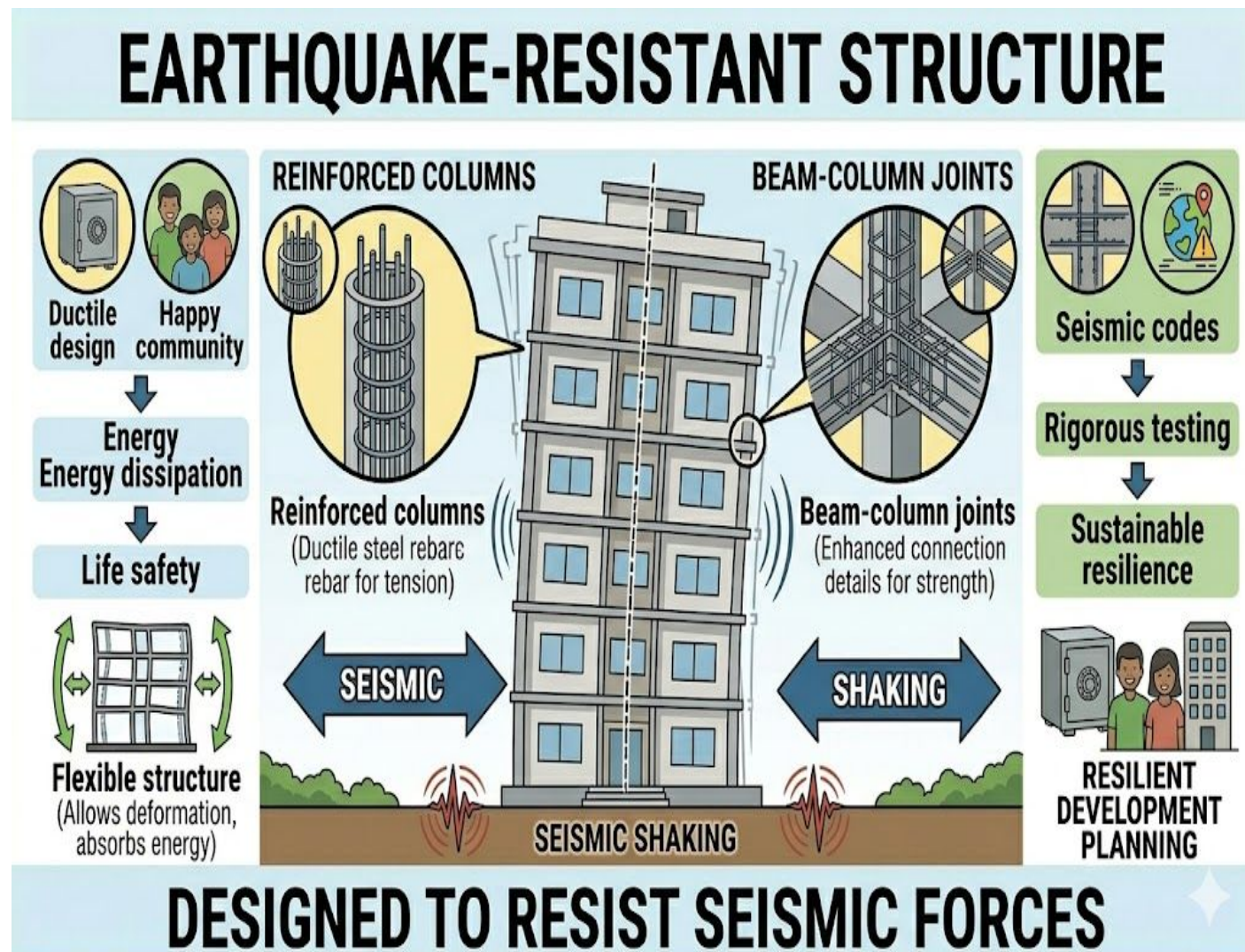
Safe Design & Construction — Concept

Safe design focuses on constructing buildings that can **withstand disaster forces**.

- It follows the principle of “**Build Back Better**”.
- Structures should be designed to **resist hazards and protect life**.
- It reduces collapse and damage during disasters.

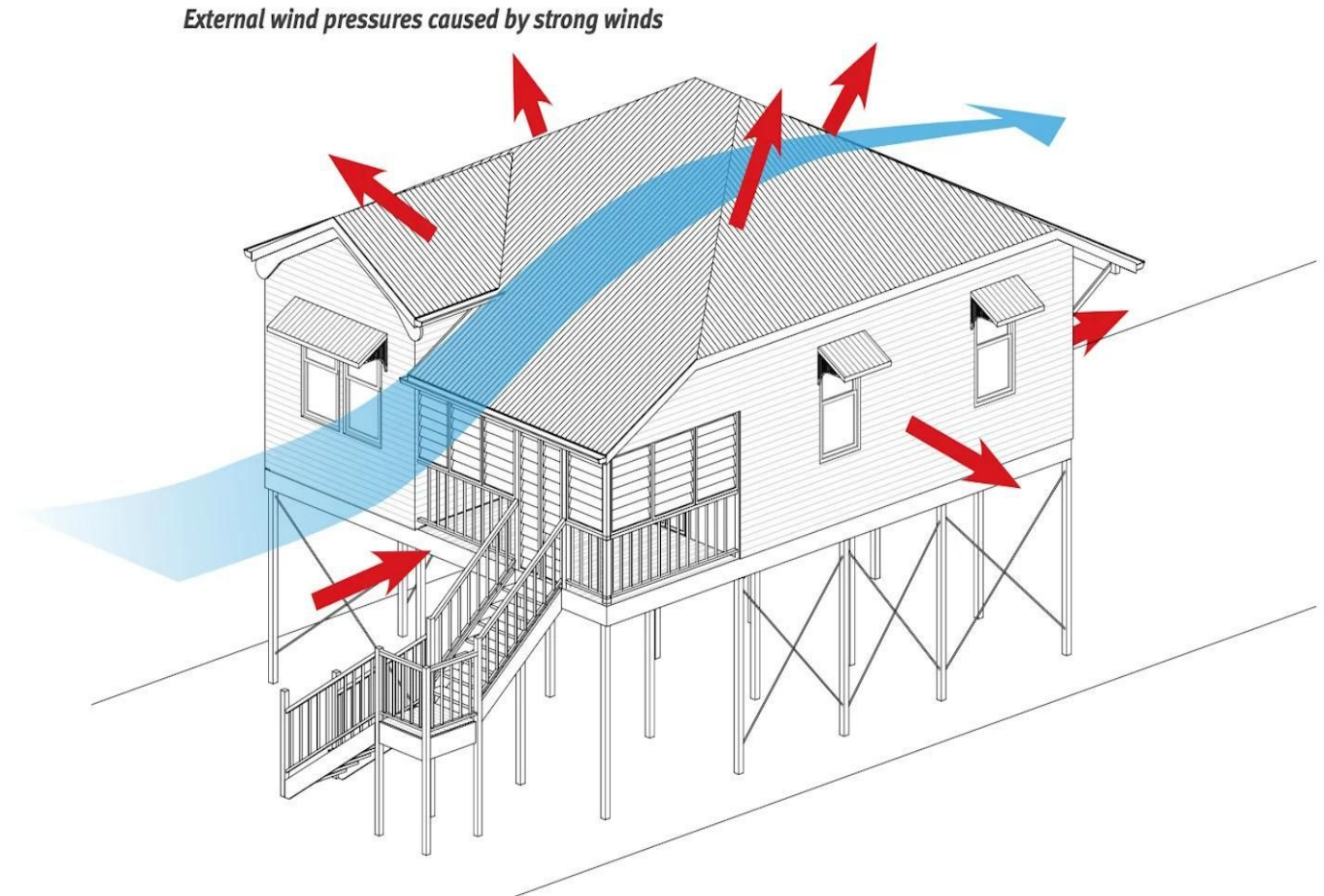


- Earthquake-resistant design ensures buildings can **withstand seismic forces**.
- Structures are designed for **ductility and flexibility**.
- Codes such as **IS 1893** and **IS 4326** are followed.
- Goal is to prevent **collapse and protect life**.



Cyclone-Resistant Design

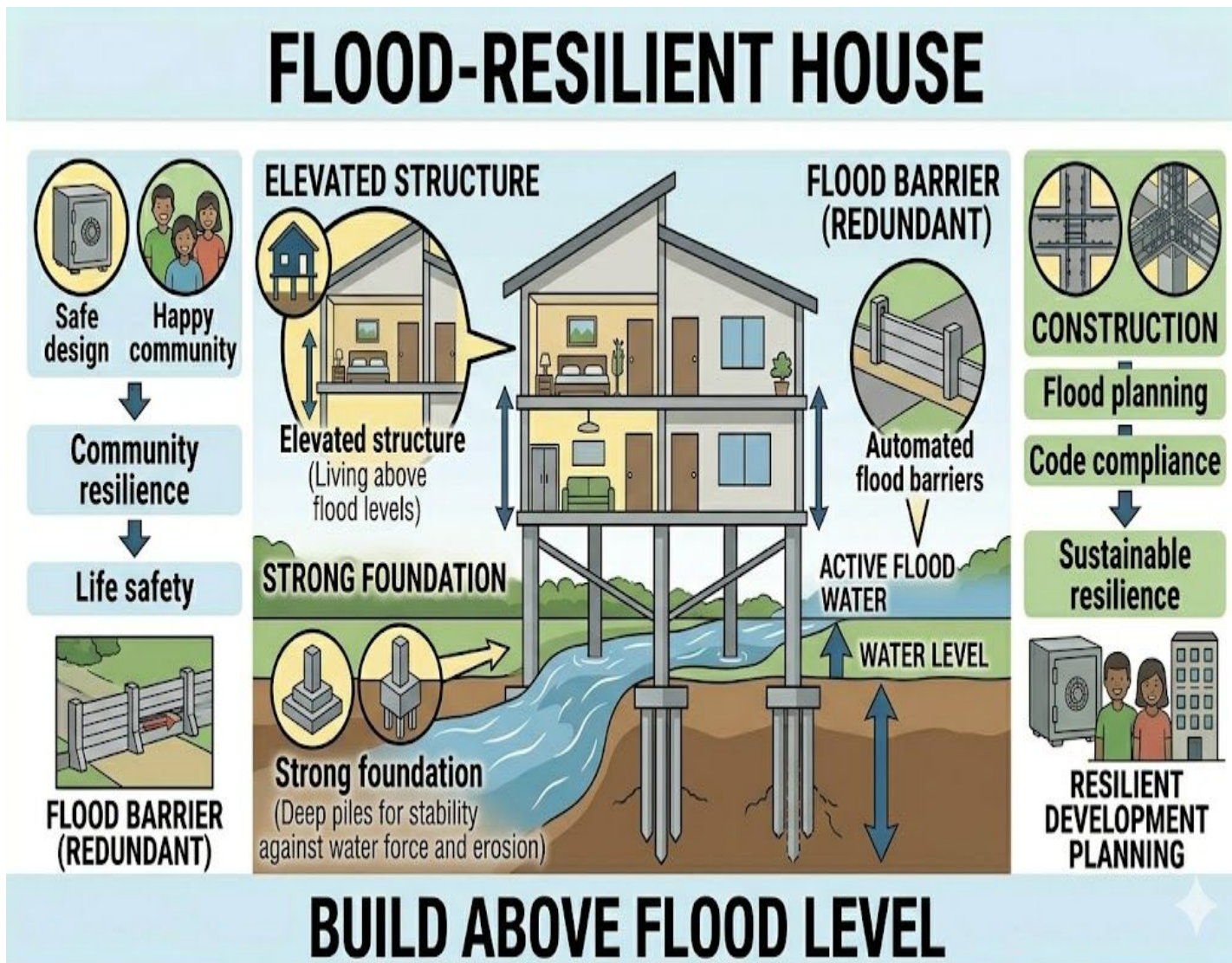
- Cyclone-resistant design protects against **high wind and storm surge**.
- Structures use **aerodynamic shapes and strong connections**.
- Roof-to-wall connections are critical for stability.
- Cyclone shelters are designed to **withstand extreme winds**.



Flood-Resilient Construction



- Flood-resistant design reduces damage from **water flow and inundation**.
- Buildings are elevated above **flood levels**.
- Water-resistant materials are used.
- Structures allow controlled water flow to reduce pressure



Structural Mitigation Measures

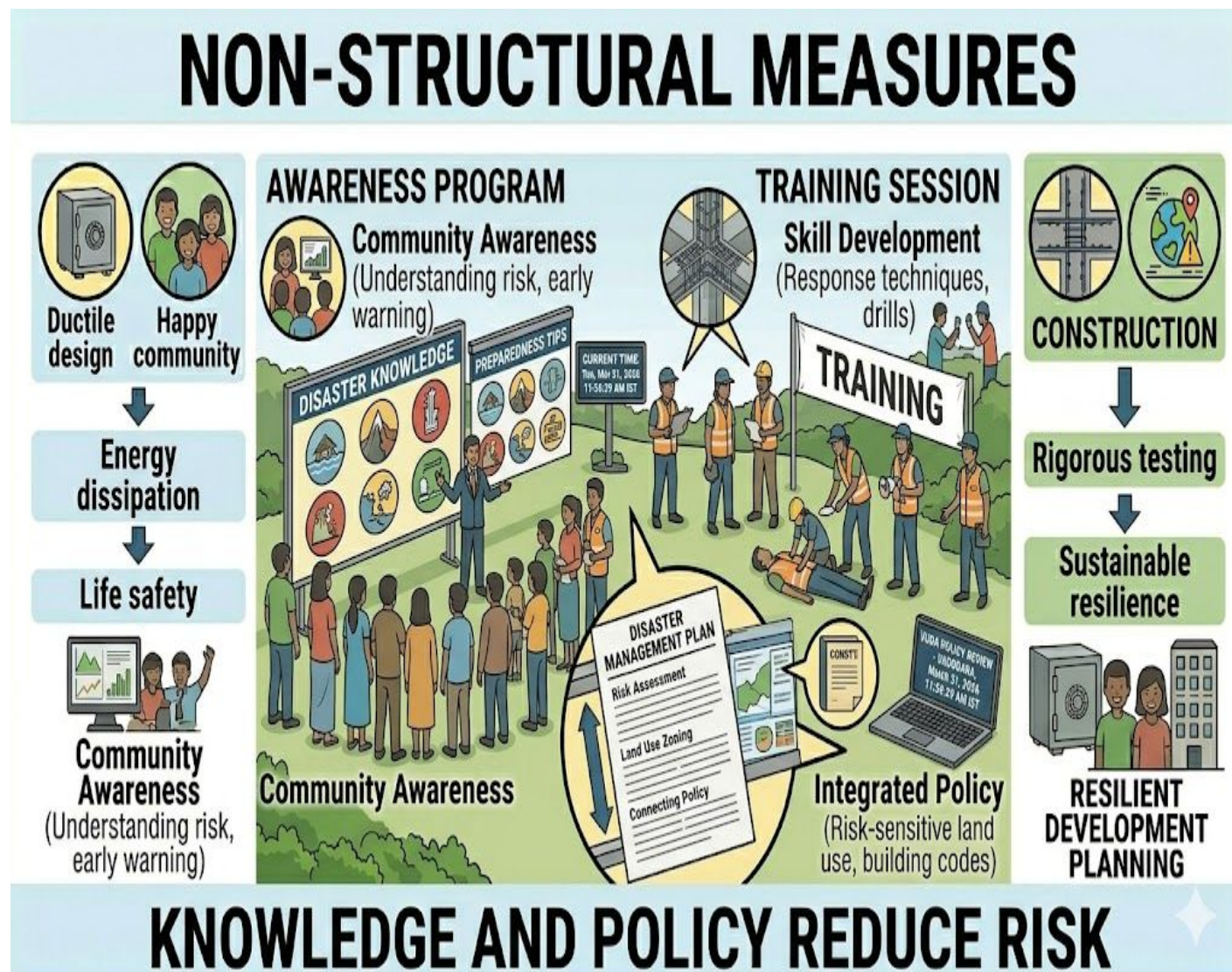
- Structural measures involve **physical constructions to reduce disaster impact.**
- Examples include **dams, embankments, cyclone shelters, and retaining walls.**
- They directly reduce the **impact of hazards on infrastructure.**
- They are essential for **large-scale protection.**



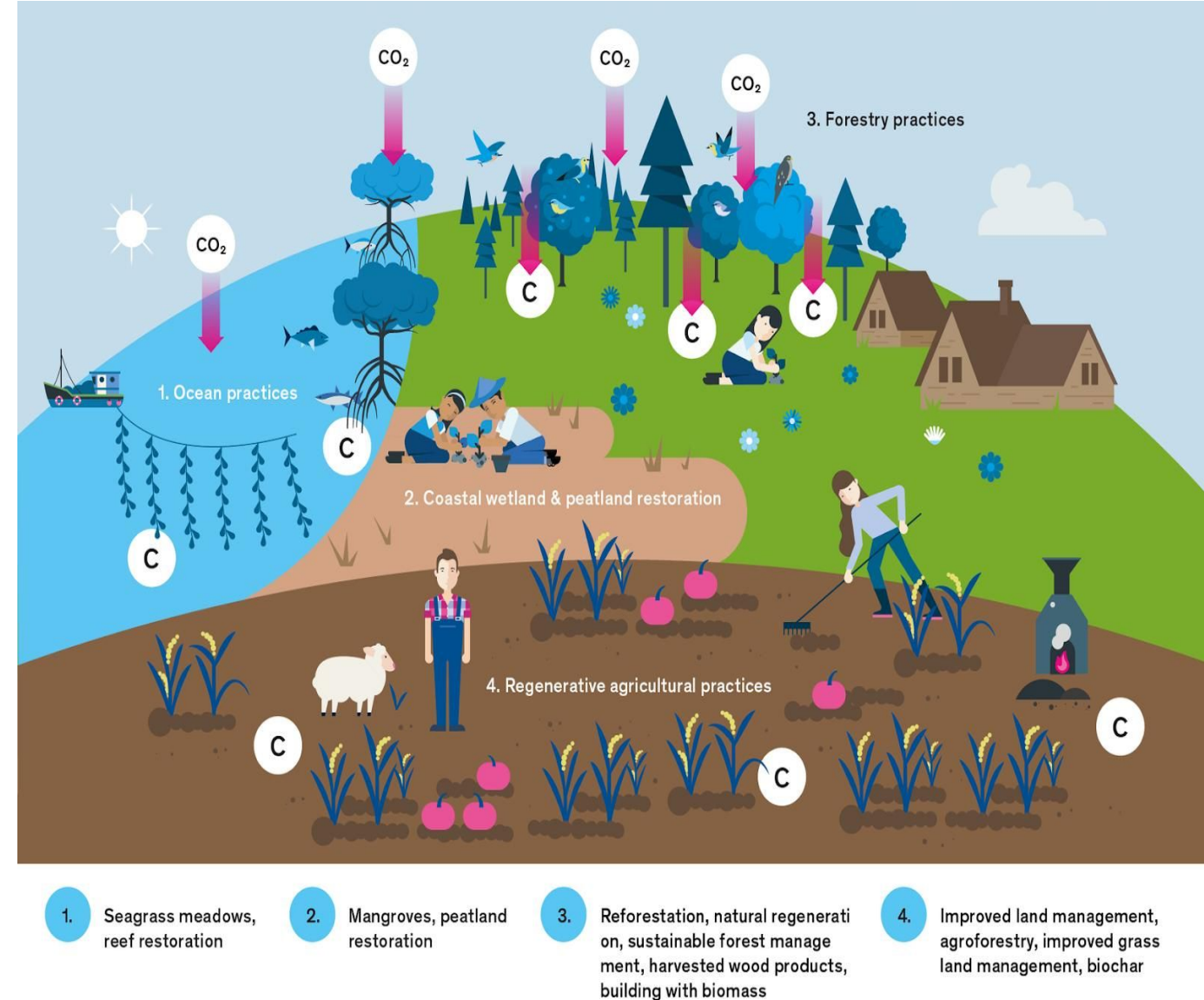
Non-Structural Mitigation Measures



- Non-structural measures include **policies, awareness, and training.**
- Examples include **building codes, land use planning, and education.**
- They are cost-effective and long-lasting.
- They complement structural measures.

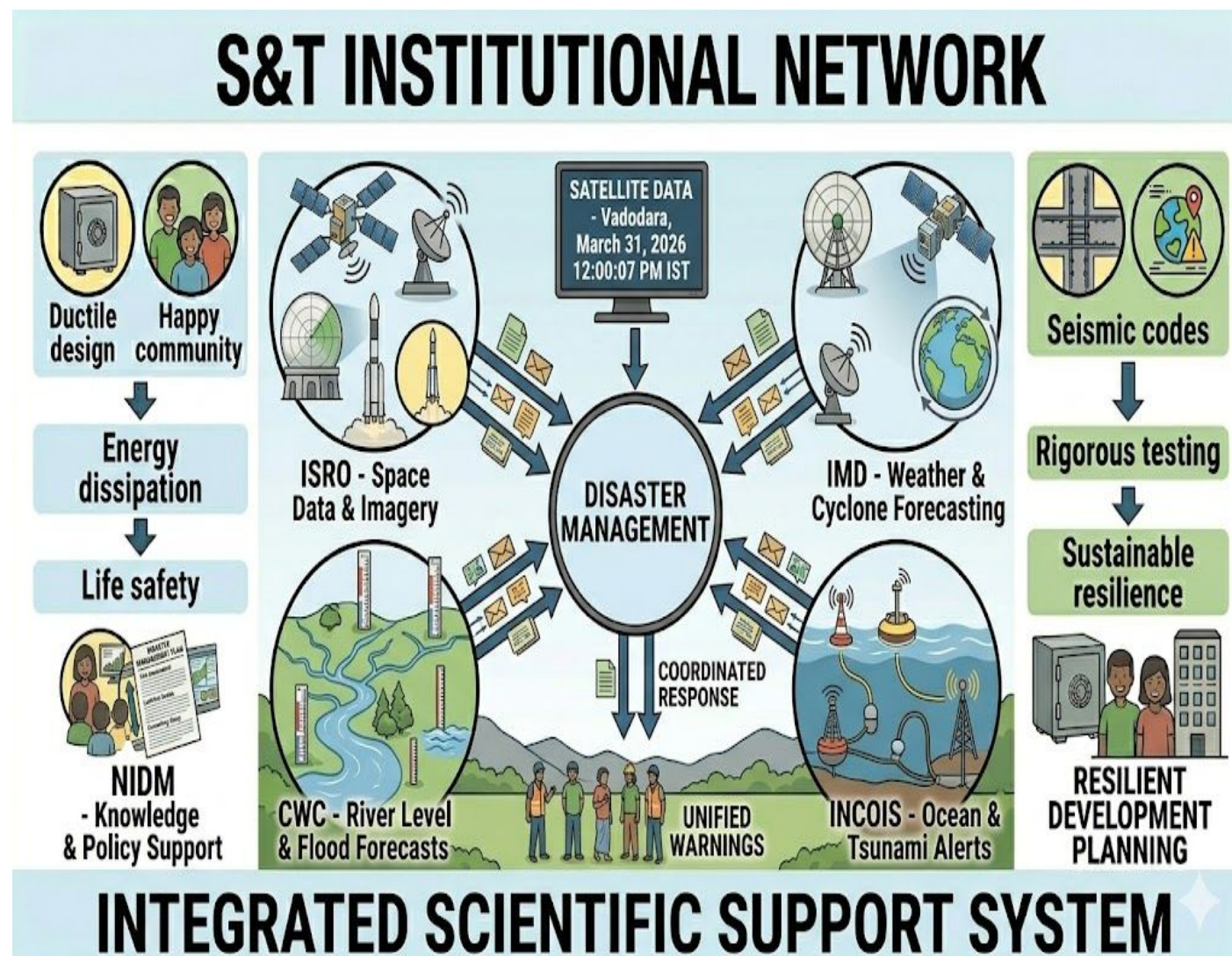


- Natural systems can reduce disaster impact.
- Wetlands absorb floodwater.
- Mangroves reduce wave energy during cyclones.
- Forests prevent landslides and soil erosion.



S&T Institutions in Disaster Management (India)

- Various institutions support disaster management using science and technology.
- ISRO and NRSC provide satellite data.
- IMD, INCOIS, and CWC provide warning systems.
- NIDM and IITs support research and training.
- These institutions work together for **effective disaster management**.





Connecting to Real-World Scenario

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Interactive Activities

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Think-Pair-Share

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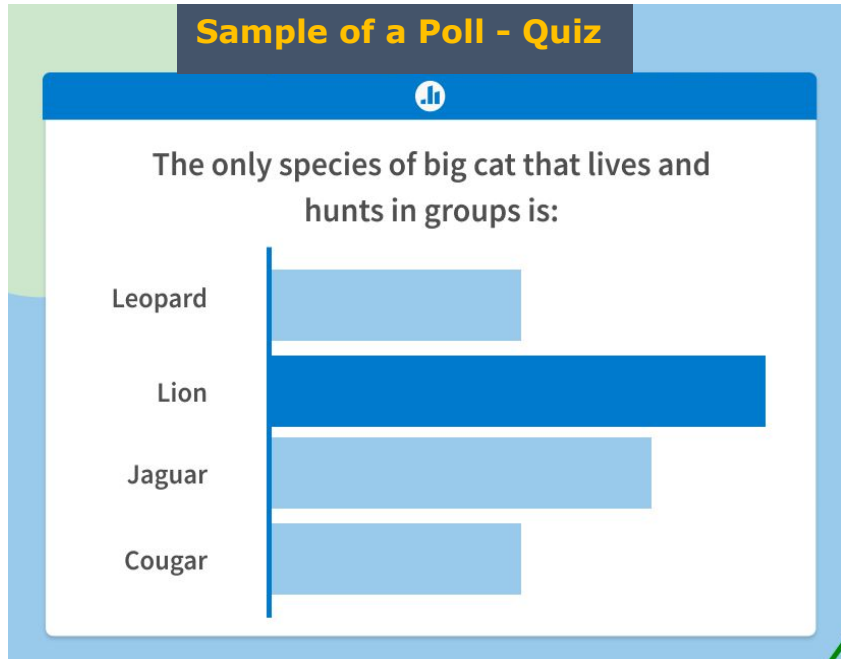
<https://i.pinimg.com/originals/9f/0e/5f/9f0e5fee28378215a04056924aca8336.jpg>

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Poll

Details of Interactive Activity



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GSFCU & Values

In every lecture, Faculty may talk on any of the following:

- **DCS & GSFCU Portal:** Remind students to use DCS, talk about it's features and instruct all to visit GSFCU website
- **Discipline, Values and Ethics.**
- **GSFC University – My University:** Emphasize the importance of taking ownership of the University.
- **Respect & Courtesy:** Remind students to be respectful to peers, teachers, and staff. Encourage polite language and behavior.
- **Punctuality:** Emphasize the importance of arriving on time to class and completing assignments by deadlines.
- **Classroom Rules:** Reinforce basic classroom rules, such as keeping mobile phones on silent and maintaining a clean environment.
- **Safety:** Remind students about safety protocols, whether it's physical safety in the classroom or digital safety when using online resources.
- **Positive Attitude:** Encourage a positive mindset, cooperation and supporting one another.

Any other thoughts related to Discipline, Values, Ethics etc. can be mentioned during these 2 minutes. You May use appropriate image / video with proper reference. One example is given in next slide.

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GSFCU & Values

Respect & Courtesy

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Dress
Appropriately



Respect
Personal Space



Be Mindful
of Your Words



Respect
Coworkers' Time



Keep Your
Workplace Clean



Socialise with
Your Peers



Use Your
Mobile Wisely



Take
Responsibility

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